

JOSHUA BARRADAS

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AWS Solutions Architect – Associate (SAA-C03) • HashiCorp Terraform Associate • AWS AI Practitioner

AWS-certified Cloud Engineer with four contract engagements and one part-time role delivering serverless integrations, multi-account security baselines, and containerised ECS Fargate platforms — Terraform, VPC design, least-privilege IAM, CI/CD and observability end-to-end. Five public AWS projects, each with live-deployment or production evidence in its repo. Full right to work in the UK.

CORE SKILLS

- **AWS:** Lambda, ECS Fargate, EC2, VPC, ALB, RDS (PostgreSQL), API Gateway, DynamoDB, S3, EventBridge, SQS/DLQ, SNS, CloudWatch, Secrets Manager, KMS, IAM, Organizations, Security Hub, GuardDuty, Textract, Cognito, OpenSearch
- **IaC & CI/CD:** Terraform (reusable modules, multi-environment), CloudFormation, GitHub Actions, policy-as-code, PR-based change control
- **Containers & Languages:** Docker, ECS Fargate, Kubernetes (GKE), Linux; Python (FastAPI, pytest, Boto3, ruff), Bash, JavaScript / Node.js
- **Networking & Security:** VPC design (subnets, route tables, security-group tiering, NAT), Route 53, Elastic Load Balancing, least-privilege IAM, KMS, Secrets Manager, OAuth 2.0
- **Architecture:** AWS Well-Architected, event-driven & serverless design, CloudWatch/X-Ray observability, cost optimisation; GCP (GKE, Terraform multi-cloud). Languages: English & Portuguese (Native)

EXPERIENCE

Cloud Engineer • Part-time — *EfficiencAI, Leeds (AI recruitment start-up)* Apr 2026 – Present • [Reference ↗](#)

- Eliminated manual token refresh and authentication failures in candidate-data sync by building a secure OAuth 2.0 JobAdder integration — 3 Lambda functions refreshing every 50 min against a 60-min token expiry (Secrets Manager, EventBridge).
- Cut environment provisioning from days to minutes by architecting a multi-cloud OnlyOffice deployment on GCP GKE with connection-based autoscaling (20+ pods), all reproducible Terraform across AWS and GCP.
- Security end-to-end: KMS-encrypted credentials, Entra ID SSO, UK data-residency, SQS dead-letter handling; real-time admin dashboard for pod health and autoscaling.
- [JobAdder integration on GitHub](#)

Cloud Engineer • Contract — *XPX Technologies, Leeds (instant-pay fintech • concurrent)* Nov 2025 – Apr 2026 • [Reference ↗](#)

- Designed a six-layer, GDPR-first reference architecture for an explainable salary-advance risk engine — synthetic-data ingestion, governed storage with lineage, offline ML with SHAP explainability, real-time scoring, and an ops layer for drift, fairness and immutable audit.
- Built a working “Score → Explain → Act” prototype of the scoring core (FastAPI / React, containerised, pytest, CI) that returns a risk band, the drivers behind it, and a recommended action — not just a number.
- [Design & prototype on GitHub](#)

Cloud Engineer • Contract — *Genyte, Leeds (precision-medicine start-up • concurrent)* Nov 2025 – Feb 2026

- Shaped compliant, well-architected designs for AI workloads in a regulated medical-data environment by communicating trade-offs on access control, environment segregation and data handling with research and engineering stakeholders.
- Removed unreviewed changes to production by establishing GitHub PR-based change control; built CloudWatch logging, metrics and alarms and supported Kubernetes deployments and pod-level debugging.

Cloud Engineer • Contract — *Pick n Pay, South Africa (major retail group)* Aug 2025 – Jan 2026

- Designed, built and piloted three AWS prototypes across two Pick n Pay stores — real-time RFID inventory (IoT Core → Lambda → DynamoDB → SAP via IDoc), finance/BI analytics (Glue, Athena, QuickSight), and real-time personalisation (API Gateway, Lambda, Amazon Personalize) — each benchmarked AWS against Azure on the same data.
- Stress-tested the RFID pipeline to a simulated five-store load (~950 msg/s at <200ms, vs ~500 msg/s on Azure) and ran controlled AWS-vs-Azure trials on throughput, latency and accuracy, measuring ~20% on-shelf inventory-accuracy improvement; tightened IAM controls under POPIA / ICASA.
- Work formed the basis of an undergraduate dissertation evaluating cloud-AI platforms for South African retail.
- [RFID inventory system on GitHub](#) [Real-time personalisation engine](#)

Cloud Engineer • Contract — *Noxieun, Leeds / Malaysia (healthcare marketing agency)* May 2025 – Aug 2025 • [Reference ↗](#)

- Reduced manual intervention and kept analytics dashboards and pipelines stable by maintaining and optimising the cloud reporting infrastructure.
- Reviewed Terraform plans pre-apply and debugged Kubernetes pod-level issues, documenting findings as reusable runbooks.

PROJECTS

All public on GitHub — each README documents exactly what was deployed and verified.

Cloud Resume Challenge *S3 · CloudFront · API Gateway · Lambda · DynamoDB · Terraform · GitHub Actions*

- Live serverless résumé site — private S3 behind CloudFront (OAC, HTTPS), visitor counter via API Gateway → Lambda → DynamoDB (atomic update), all in Terraform with remote state, and GitHub Actions CI/CD authenticating via OIDC (no stored keys). Lighthouse 100/100/100/100, 5 passing tests. [live site](#) · [code](#)

ECS Fargate VPC Platform *FastAPI · ECS Fargate · Terraform · RDS · ALB · Docker*

- Containerised REST API in a custom multi-AZ VPC (45+ resources) with three-tier security groups and RDS PostgreSQL, all in Terraform; deployed live and verified. 18 tests, CI, and a Boto3 cost-audit tool that flagged the NAT gateway (~\$32/mo) as top cost driver. [code + deploy evidence](#)

Secure Multi-Account AWS Landing Zone *Terraform · AWS Organizations · IAM Identity Center*

- Reusable security baseline — 64 Terraform resources across 6 modules (SCP guardrails, least-privilege IAM, Security Hub + GuardDuty, policy-as-code CI). Security stack deployed live to AWS and verified; Organizations and Identity Center layers written and validated (not deployed). [code + deploy evidence](#)

JobAdder OAuth AWS *Lambda · CloudFormation · KMS · Secrets Manager · SQS/DLQ · SNS*

- Serverless OAuth 2.0 token-lifecycle integration — 3 Lambda functions refreshing every 50 min against a 60-min expiry (zero lapses), with DLQ and SNS alerting; sanitised reference of a system running in production on AWS for EfficiencyAI. 9 unit tests. [live demo](#) · [code](#)

IntelliDoc Engine *Textract · Lambda · Terraform · DynamoDB · OpenSearch · Cognito*

- Serverless receipt/invoice pipeline with full-text search behind a Cognito-authenticated API Gateway; deployed live and verified (us-east-1) — 565ms response in testing, 100% success in testing. 23 passing pytest tests. [code + evidence](#)

EDUCATION & CERTIFICATIONS

BA (Hons) Business Management with Marketing — University of Leeds

Graduated July 2025

- AWS Solutions Architect – Associate (SAA-C03) · HashiCorp Terraform Associate (004) · AWS AI Practitioner · Microsoft Applied Skills (Azure): Knowledge Mining, AI Agent, Generative AI
- Toastmasters (Leeds City) and White Rose Speakers — regular public speaking and presentation practice.